In[ ]:=
```
n1 = 3;
g1 = 3;
n2 = 3;
g2 = 1;
n3 = 3;
g3 = 1;


(*Consistencies solution for the choice of pentaquark *)

γ12bar = 0.3954232044351274; (* relationship between light and anti-charm quark *)
β12 =  0.22559600009756925; (* relationsip between light and charm quark *)
β22bar = 0.859355921116699`; (* relationship between charm and anti-charm quark *)

(*Defining Constants*)

xx = 0.346;
(*This is the value of αs which is αs=g^2/(ħc),
used from the mesonic fit as done in meson calculations.*)
alfas = xx;
(*aa=.178;*)
 zz = 306.;(*This is the mass of light quark like up, down*)
 aa = 0.197; (*This is the ratio of light mass to the heavy mass of quarks *)
yy = 107.6;
B0 = yy; (*Bag Constant *)
a1 = 1 / zz; (*This is α=ħ/(m1 c), where m1 is the mass of light quark. here,
reduced planck's constant and c both are set to be 1.*)
P = N[Pi, 7];
alfa1 = 1.;(* Ratio of mass of light quarks, BA, 13 September *)
alfa2 = aa; (* Ratio of mass of light quark to the heavy quark,
BA, 13 September *)

η = (n1 * g1 + n3 * g3 + n2 * g2) / 5;
theta = -6 η / (5 (5 η - 1)); (*Just a multiplication factor for a particular value of eta *)
Q1 = -0.5142857142857142; (*Q1 is the constant in the differential equation  d^2 f1/dx^2 =
```

Q1 $\frac{f1^{\frac{3}{2}}}{\sqrt{x}}$ for the outer region*)

```
Q3 = -0.973694508123489;
(*Q3 is the constant in the differential equation  d^2 f3/dx^2 =
 Q3 f3^(3/2)/Sqrt[x] in the middle region. This is obtained by solving two
  consistency conditions along with other constant γ12bar. *)
Q2 = -2.007183585486792;
(*Qbar is the constant in the differential equation  d^2 f3/dx^2 =
 Q2 f3^(3/2)/Sqrt[x] in the inner region. This is obtained by solving three
    consistency conditions along with other constants β13 & β23. *)
g = alfas;(*Why  g = alfas in this case ?*)
(* I think there should be g = /sqrt{alfas} in order to satisfy the multiplication
  facor in the expreesion of kinetic energy i.e. 'c', BA, 13 September *)
R = 3 a/(8 * alfas) * ((3 π η)/2)^(2/3);(*Page number 54, it is inverse in the copy though *)
c = 12/(5 π) 4/3 g^2 - 4/3 alfas/a ((3 π η)/2)^(1/3);
(*Multiplication factor in the expression of kinetic energy *)
d = 4/3 g^2 - 4/3 alfas/a ((3 π η)/2)^(1/3) 18/(25 π) η/(η - 1/5); (* 18g^2/(5(5η -1 )).16 π^2.(3a/8alfas). R^2/(6π^2)^2 ,
The value of R is from page number 54 which is written inverse here,
and (3a/8alfas) is inverse in copy, BA, 13 September *)
(*Multiplication factor in the expression of Potential energy *)


(*Let us define the three functions for three regions *)

f3[x3_?NumericQ, x2_?NumericQ, u2_?NumericQ , s2_?NumericQ] :=
  Abs[x3 * y3'[x3] - y3[x3] - n3 * Q3/(3 η) - β22bar * n2 * Q2/(3 η) + β22bar^(3/2) * n2 * Q3/(3 η)]
```

```
(*Appendix from Kinetic energy*) /.
 First@NDSolve[{y3''[x] == (Q3 * Abs[y3[x]^(3/2)] / x^(1/2)), y3'[x2] == β22bar * s2 ,
    y3[x2] == β22bar * u2 (* It shows how the function f3
     (x3) changes on other regions*)}, y3, {x, x2, x3}];
(*s2 is the first derivative of function f_2(x) *)
(* u2 is the function f_2(x) *)
(*This is the function f3 defined on the
  middle region. It has the differential equation  d^2 f3/dx^2 =
 Q3 f3^(3/2)/Sqrt[x] in for x3<x<x2 which is middle region. On this region, f1(x)=γ12 f2(x).  *)
f1[x1_?NumericQ, x3_?NumericQ, s3_?NumericQ, u3_?NumericQ] :=
  Abs[x1 * y1'[x1] - y1[x1] - n1 * Q1/(3 η) - γ12bar * n3 * Q3/(3 η) +
    γ12bar^(3/2) * n3 * Q1/(3 η) - (γ12bar^(3/2) * β22bar^(3/2) - β12^(3/2)) * n2 * Q1/(3 η) +
    γ12bar * β22bar^(3/2) * n2 * Q3/(3 η) - γ12bar * β22bar * n2 * Q2/(3 η)]
   (*Appendix from Kinetic energy*) /. First@NDSolve[{y1''[x] == (Q1 * Abs[y1[x]^(3/2)] / x^(1/2)),
     y1'[x3] == γ12bar * s3, (* s3 is the first derivative of function f_2bar(x) *)
     (* It means at the boundary x3 the function df1 changes to other function *)
     y1[x3] == γ12bar * u3 (* u3 is the function f_2bar(x) *)
     (*It means at the boundary x3 the function f1 changes to other function*)
    }, (* It means at the boundary x3 the function f1 changes to other function *)
    y1, {x, x3, x1}]; (*This is the function defined in outer region for x2<
  x<x1 so called f1. It has differential equation  d^2 f1/dx^2 =
 Q1 f1^(3/2)/Sqrt[x] for the outer region which is solved by using boundary
    condition of (d/dx) f1(x)|_x2 = γ12bar (d/dx) f2(x)|_x2 and f1(x3)=γ12 f2(x3)  *)
```

```
f2[xmax_?NumericQ, m2_?NumericQ] :=
  Module[{xx = xmax},
   solution2 = NDSolve[
     {y2''[x] == (Q2 * Abs[y2[x]^(3/2)]) / x^(1/2), y2'[1.*^-9] == m2, y2[1.*^-9] == 0.}, y2, {x, 1.*^-9, xx}];
   m = NIntegrate[Sqrt[x] * Evaluate[y2[x] /. solution2]^(3/2),
     {x, 1.*^-9, xx}, Method → {Automatic, "SymbolicProcessing" → 0}];
   z = m[[1]];
   z = z - n2 / (3. * η);
   (*z=z-n/(A)*);
   Return[Abs[z]]; Remove["NDSolve`*$*$1"]];
(*This is the function f3 defined on the inner region. It has the differential
  equation  d^2 f3/dx^2 = Q3 f3^(3/2)/Sqrt[x]  for 0<x<x_3 inner region. On this region,
f_1(x)=β_13 f_3(x) and f_2(x)= β_23 f_3(x).    *)

(*Start the program*)

m2 = 22;
For[j = 1., j ≤ 1., j++, B = (B0)^4;
  a = a1; (* It must be to satisfy the multiplication
   factor used in kinetic energy i.e. c, BA, 13 September *)
  (*Now for the first pentaquark,
  eta=1 which in total contains 5 quarks which can be written by *)

 mqn = If[j > 1, m2 + 2, m2]; (*case2new*)(*mqn=If[j>1,m2+4,m2];*)
  m2 = mqn;
  topi = 0.01;
  m2B = m2;
  Energy = 10^9;
  Energy = 10^9;
  Energy2plus = 10^9;
```

```
H = 10^9 + 1;
SN = 0;
nloop = 1;
For[n = 1, n ≤ 3, n++, While[Energy ≤ H, F = H;
  H = Energy;
  HX = Energy6bare;
  HY = Energy2plus; m2 = m2 + SN * 10^(-n);
  x2 = 2.*^-8; (*Print[m2];*)(*was 0 .5 1.66*)
  FindMinimum[f2[x2, m2], {x2, topi}];
  (*remove this function and investigate if it is actually required BS 4/3/2021*)
  x2 = x2 /. Last[FindMinimum[f2[x2, m2], {x2, topi}]];
  solution2 = NDSolve[{y2''[x] == (Q2 * Abs[y2[x]^(3/2)]) / x^(1/2),
     y2'[1.*^-9] == m2, y2[1.*^-9] == 0.}, y2, {x, 1.*^-9, x2}];
  t1 = Evaluate[y2[x2] /. solution2];
  t4 = Evaluate[y2'[x2] /. solution2];
  u2 = t1[[1]]; (* u2 is the function f_2(x) *)
  s2 = t4[[1]]; (* s2 is the first derivative of function f_2(x) *)
  If[u2 < 0., u2 = 0.];


  x3 = x2 + 0.1;
  FindMinimum[{f3[x3, x2, u2, s2]}, {x3, x2}];
  x3 = x3 /. Last[FindMinimum[{f3[x3, x2, u2, s2]}, {x3, x2}]];
  solution3 = NDSolve[{y3''[x] == (Q3 * Abs[y3[x]^(3/2)]) / x^(1/2),
     y3'[x2] == β22bar * s2 , y3[x2] == β22bar * u2}, y3, {x, x2, x3}];
  t2 = Evaluate[y3[x3] /. solution3];
  t5 = Evaluate[y3'[x3] /. solution3];
  u3 = t2[[1]]; (* u3 is the function f_2bar(x) *)
  s3 = t5[[1]]; (* s3 is the first derivative of function f_2bar(x) *)
  If[u3 < 0., u3 = 0.];

  x1 = x3 + 0.1;
  FindMinimum[{f1[x1, x3, s3, u3]}, {x1, x3}];
  x1 = x1 /. Last[FindMinimum[{f1[x1, x3, s3, u3]}, {x1, x3}]];
```

```
solution1 = NDSolve[{y1''[x] == (Q1 * Abs[y1[x]^(3/2)] / x^(1/2)),
    y1'[x3] == γ12bar * s3 , y1[x3] == γ12bar * u3 }, y1, {x, x3, x1}];
t3 = Evaluate[y1[x1] /. solution1];
t6 = Evaluate[y1'[x1] /. solution1];
u1 = t3[[1]]; (* u1 is the function f_1(x) *)
s1 = t6[[1]]; (* s1 is the first derivative of function f_1(x) *)
If[u1 < 0., u1 = 0.];


(*Integrations *)(* These integrations change values but outer and outer 1
 doesnot because they depend on x and derivates of x (BA, 11 September)*)

normmx2 = x2 s2 - u2 ;
normx2 = n2 Q2 / (3 η) ;
(*normx2barr and normx2bar should be equal *)
normmx3 = x3 s3 - u3 ;
normx3 = n3 Q3 / (3 η) + β22bar n2 Q2 / (3 η) - (β22bar)^(3/2) n2 Q3 / (3 η);
(*normmx2 and normx2 should be equal *)
normmx1 = x1 s1 - u1 ;
normx1 = n1 Q1 / (3 η) + γ12bar n3 Q3 / (3 η) - (γ12bar)^(3/2) n3 Q1 / (3 η) +
  ((γ12bar)^(3/2) (β22bar)^(3/2) - (β12)^(3/2)) n2 Q1 / (3 η) - γ12bar (β22bar)^(3/2) n2 Q3 / (3 η) + γ12bar β22bar n2 Q2 / (3 η) ;
(*normmx1 and normx1 should be equal *)

h1 = NIntegrate[Evaluate[y2[x] /. solution2]^2.5 / (x^0.5),
  {x, 0, x2}, Method → {Automatic, "SymbolicProcessing" → 0}];
h2 = NIntegrate[Evaluate[y3[x] /. solution3]^2.5 / (x^0.5), {x, x2, x3},
  Method → {Automatic, "SymbolicProcessing" → 0}];
h3 = NIntegrate[Evaluate[y1[x] /. solution1]^2.5 / (x^0.5), {x, x3, x1},
  Method → {Automatic, "SymbolicProcessing" → 0}];
h4 = h1[[1]];
h5 = h2[[1]];
```

```
h6 = h3[[1]] ;
AB = 4/7 Sqrt[x2] (u2)^(5/2) - 5 n2/(21 η) s2 ;
CD = 4/7 Sqrt[x3] (u3)^(5/2) - 4/7 (β22bar)^(5/2) Sqrt[x2] (u2)^(5/2) -
   5/(7 Q3) s3 (n3 Q3/(3 η) + β22bar n2 Q2/(3 η) - (β22bar)^(3/2) n2 Q3/(3 η)) + 5/(7 Q3) (β22bar)^2 n2 Q2/(3 η) s2 ;
CCD = 4/7 Sqrt[x3] (u3)^(5/2) - 4/7 Sqrt[x2] (β22bar)^(5/2) (u2)^(5/2) - 5/(7 Q3) s3 normmx3 +
   5/(7 Q3) (β22bar)^2 normmx2 s2 ;

EF = 4/7 Sqrt[x1] (u1)^(5/2) - 4/7 (γ12bar)^(5/2) Sqrt[x3] (u3)^(5/2) -
   5/(7 Q1) s1 (n1 Q1/(3 η) + γ12bar n3 Q3/(3 η) - (γ12bar)^(3/2) n3 Q1/(3 η) + n2 Q1/(3 η) ((γ12bar)^(3/2) (β22bar)^(3/2) - (β12)^(3/2)) -
      (γ12bar) (β22bar)^(3/2) n2 Q3/(3 η) + (γ12bar) (β22bar) n2 Q2/(3 η)) +
   5/(7 Q1) (γ12bar)^2 s3 (n3 Q3/(3 η) + β22bar n2 Q2/(3 η) - (β22bar)^(3/2) n2 Q3/(3 η)) ;
(* h4 and AB should be equal *)
(* h5 and CD should be equal *)
(* h6 and EF should be equal *)

K222 = n2/(3 η Q2) s2 - 1/Q2 h4 ;
K22 = 4 n2/(7 η Q2) s2 - 4/(7 Q2) Sqrt[x2] (u2)^(5/2) ;
(*K2bar2barr and K2bar2bar should be equal *)
K32 = - 4/(7 Q2) β22bar^1.5 Sqrt[x2] (u2)^(5/2) +
```

$$\frac{n2}{3\,\eta}\left(\frac{\beta22bar^{1.5}}{Q2}-\frac{\beta22bar}{Q3}+\frac{5*\beta22bar^{1.5}}{7\,Q2}\right)s2+\frac{n2}{3\,\eta\,Q3}s3\,;$$

$$K332=\frac{n2\,(\beta22bar)^{\frac{3}{2}}}{3\,\eta\,Q2}s2-\frac{n2\,\beta22bar}{3\,\eta\,Q3}s2+\frac{n2}{3\,\eta\,Q3}s3-\frac{(\beta22bar)^{\frac{3}{2}}}{Q2}h4;$$

(*K2bar2 and K2barr2 should be equal *)

$$K21=-\frac{4}{7\,Q2}\beta12^{1.5}\sqrt{x2}\;(u2)^{\frac{5}{2}}+\frac{n2}{3\,\eta}\left(\frac{5}{7\,Q2}\beta12^{1.5}+\frac{\beta12^{1.5}}{Q2}-\frac{\gamma12bar^{1.5}*\beta22bar}{Q3}\right)s2+$$

$$\frac{n2}{3\,\eta}\left(\frac{\gamma12bar^{1.5}}{Q3}-\frac{\gamma12bar}{Q1}\right)s3+\frac{n2}{3\,\eta\,Q1}s1\,;$$

$$K221=\frac{n2}{3\,\eta\,Q2}(\beta12)^{\frac{3}{2}}s2+\frac{n2\,(\gamma12bar)^{\frac{3}{2}}}{3\,\eta\,Q3}s3-$$

$$\frac{n2}{3\,\eta\,Q3}(\gamma12bar)^{\frac{3}{2}}\beta22bar\,s2+\frac{n2}{3\,\eta\,Q1}s1-\frac{n2}{3\,\eta\,Q1}\gamma12bar\,s3-\frac{(\beta12)^{\frac{3}{2}}}{Q2}h4\,;$$

(*K2bar1 and K2barr1 should be equal *)

$$K31=\frac{4}{7}\beta22bar^{1.5}\left(\frac{\gamma12bar^{1.5}*\beta22bar}{Q3}-\frac{\beta12^{1.5}}{Q2}\right)\sqrt{x2}\;(u2)^{\frac{5}{2}}-\frac{4}{7\,Q3}\gamma12bar^{1.5}\sqrt{x3}\;(u3)^{\frac{5}{2}}+$$

$$\frac{n2}{3\,\eta}\beta22bar\left(\frac{5}{7\,Q2}\beta22bar^{\frac{1}{2}}*\beta12^{1.5}-\frac{\gamma12bar^{1.5}*\beta22bar^{1.5}}{Q3}-\frac{5\,Q2}{7\,Q3^2}\gamma12bar^{1.5}*\right.$$

$$\left.\beta22bar+\frac{Q2}{Q3^2}\gamma12bar^{1.5}*\beta22bar+\frac{1}{Q2}\beta12^{1.5}*\beta22bar^{\frac{1}{2}}-\frac{\beta12^{1.5}}{Q3}\right)s2+$$

$$\frac{1}{3\,\eta\,Q3}\left(n3*\gamma12bar^{1.5}-\frac{n3\,Q3}{Q1}\gamma12bar+n2*\beta12^{1.5}-\frac{n2\,Q2}{Q3}\gamma12bar^{1.5}*\beta22bar+\right.$$

$$\left.\frac{5}{7}\gamma12bar^{1.5}\left(n3+\frac{n2\,Q2}{Q3}\beta22bar-n2*\beta22bar^{1.5}\right)\right)s3+\frac{n3}{3\,\eta\,Q1}s1\,;$$

$$(*K122 = \frac{n2}{3\,\eta}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}\left[\frac{1}{Q3}s3 - \frac{\beta 22bar}{Q3}s2\right] - \frac{(\beta 12)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}}{Q2}h4 - \frac{(\gamma 12bar)^{\frac{3}{2}}}{Q3}h5$$

$$- \frac{n2\,Q2}{3\,\eta\,Q3}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\left[\frac{1}{Q3}s3 - \frac{\beta 22bar}{Q3}s2\right] + \frac{n3}{3\,\eta\,Q1}s1 - \frac{n3}{3\,\eta\,Q1}\gamma 12bar\,s3$$

$$+ (\beta 12)^{\frac{3}{2}}\frac{n2}{3\,\eta}\left[\frac{(\beta 22bar)^{\frac{3}{2}}}{Q2}s2 + \frac{1}{Q3}s3 - \frac{\beta 22bar}{Q3}s2\right] +$$

$$\frac{n3}{3\,\eta\,Q3}(\gamma 12bar)^{\frac{3}{2}}s3 - \frac{n2}{3\,\eta\,Q3}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}s3\ ;\ *)$$

$$K331 = \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}s3 - \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}\beta 22bar\,s2 - \frac{(\beta 12)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}}}{Q2}h4 - \frac{(\gamma 12bar)^{\frac{3}{2}}}{Q3}h5 -$$

$$\frac{n2\,Q2}{3\,\eta\,Q3^2}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\,s3 + \frac{n2\,Q2}{3\,\eta\,Q3^2}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^2\,s2 + \frac{n2}{3\,\eta}(\beta 22bar)^{\frac{3}{2}}$$

$$\left(\frac{\beta 12^{\frac{3}{2}}}{Q2} - \frac{(\gamma 12bar)^{\frac{3}{2}}\beta 22bar}{Q3}\right)s2 + \frac{n3}{3\,\eta}\gamma 12bar\left(\frac{\gamma 12bar^{\frac{1}{2}}}{Q3} - \frac{1}{Q1}\right)s3 + \frac{n3}{3\,\eta\,Q1}s1\ ;$$

(*K12 and K122 should be equal*)

$$K33 = \frac{4}{7}\beta 22bar^2\left(\frac{\beta 22bar^{\frac{1}{2}}}{Q3} - \frac{\beta 22bar}{Q2}\right)\sqrt{x2}\,(u2)^{\frac{5}{2}} -$$

$$\frac{4}{7\,Q3}\sqrt{x3}\,(u3)^{\frac{5}{2}} + \frac{n2}{3\,\eta}\beta 22bar^2\left(\beta 22bar\frac{12}{7\,Q2} + \frac{2}{7}\frac{Q2}{Q3^2} - \frac{2*\beta 22bar^{\frac{1}{2}}}{Q3}\right)s2 +$$

$$\left(\frac{12\,n3}{21\,\eta\,Q3} + \beta 22bar\frac{n2}{3\,\eta\,Q3}\left(\beta 22bar^{\frac{1}{2}} - \frac{2}{7}\frac{Q2}{Q3} - \frac{5}{7}\beta 22bar^{\frac{1}{2}}\right)\right)s3\ ;$$

$$K333 = \frac{n2\,Q2}{3\,\eta\,Q3}\left(\frac{(\beta 22bar)^{\frac{3}{2}}}{Q2} - \frac{\beta 22bar}{Q3}\right)s3 - \frac{n2\,Q2\,\beta 22bar}{3\,\eta\,Q3}\left(\frac{(\beta 22bar)^{\frac{3}{2}}}{Q2} - \frac{\beta 22bar}{Q3}\right)s2 -$$

$$\frac{(\beta 22bar)^3}{Q2}h4 - \frac{1}{Q3}h5 + (\beta 22bar)^3\frac{n2}{3\,\eta\,Q2}s2 - (\beta 22bar)^{\frac{5}{2}}\frac{n2}{3\,\eta\,Q3}s2 + \frac{n3}{3\,\eta\,Q3}s3\ ;$$

(*K22 and K222 should be equal *)

(*THese Y and Z are to check the value of integral of power $\frac{5}{2}$ in the outer region. Which proved to be false yet. I need to work on it extensively.*)

$$K11 = \frac{4}{7}\left(\frac{\beta22bar^{\frac{5}{2}} * \gamma12bar^3}{Q3} - \frac{\beta12^3}{Q2}\right)\sqrt{x2}\ (u2)^{\frac{5}{2}} + \frac{4}{7}\left(\frac{\gamma12bar^{\frac{5}{2}}}{Q1} - \frac{\gamma12bar^3}{Q3}\right)\sqrt{x3}\ (u3)^{\frac{5}{2}} -$$

$$\frac{4}{7\,Q1}\sqrt{x1}\ (u1)^{\frac{5}{2}} + \left(\frac{n2}{3\,\eta}\left(\frac{\beta12^3}{Q2} - \frac{\beta12^{\frac{3}{2}} * \gamma12bar^{\frac{3}{2}} * \beta22bar}{Q3} + \frac{Q2}{Q3^2}\gamma12bar^3 *\right.\right.$$

$$\left.\beta22bar^2 - \frac{\beta12^{1.5} * \gamma12bar^{1.5} * \beta22bar}{Q3}\right) +$$

$$\left.\frac{5\,n2}{21\,\eta}\left(\frac{\beta12^3}{Q2} - \frac{Q2}{Q3^2}\beta22bar^2 * \gamma12bar^3\right)\right)s2 +$$

$$\left(\frac{n2}{3\,\eta}\left(\frac{\beta12^{1.5} * \gamma12bar^{1.5}}{Q3} - \frac{\beta12^{1.5} * \gamma12bar}{Q1} - \frac{\beta22bar^{1.5} * \gamma12bar^3}{Q3} +\right.\right.$$

$$\frac{\beta22bar^{1.5} * \gamma12bar^{\frac{5}{2}}}{Q1} + \frac{\beta12^{1.5} * \gamma12bar^{1.5}}{Q3} - \frac{Q2}{Q3^2}\gamma12bar^3 * \beta22bar -$$

$$\frac{\beta12^{1.5} * \gamma12bar}{Q1} + \frac{\gamma12bar^{\frac{5}{2}} * \beta22bar^{1.5}}{Q1} + \frac{Q2}{Q1^2}\gamma12bar^2 * \beta22bar -$$

$$\left.\left.\frac{Q3}{Q1^2}\beta22bar^{1.5} * \gamma12bar^2\right) + \frac{n3}{3\,\eta}\left(\frac{\gamma12bar^3}{Q3} - \frac{2 * \gamma12bar^{\frac{5}{2}}}{Q1} + \frac{Q3}{Q1^2}\gamma12bar^2\right) +\right.$$

$$\frac{5}{21\,\eta}\left(\frac{n3}{Q3}\gamma12bar^3 + \frac{n2\ Q2}{Q3^2}\gamma12bar^3 * \beta22bar - \frac{n2}{Q3}\beta22bar^{1.5} * \gamma12bar^3 -\right.$$

$$\left.\left.\frac{n3\ Q3}{Q1^2}\gamma12bar^2 - \frac{n2\ Q2}{Q1^2}\gamma12bar^2 * \beta22bar + \frac{n2\ Q3}{Q1^2}\gamma12bar^2 * \beta22bar^{1.5}\right)\right) *$$

$$s3 + \left(\frac{n1}{3\,\eta\,Q1} - \frac{n3\,Q3}{3\,\eta\,Q1^2}\gamma12bar + \frac{n3}{3\,\eta\,Q1}\gamma12bar^{1.5} + \frac{n2}{3\,\eta\ Q1}\right.$$

$$\left(\beta 12^{1.5} - \gamma 12bar^{1.5} * \beta 22bar^{1.5} - \frac{Q2}{Q1}\beta 22bar * \gamma 12bar + \frac{Q3}{Q1}\beta 22bar^{1.5} * \gamma 12bar\right) + \frac{5}{21\,\eta}\left(\frac{n1}{Q1} + \frac{n3\ Q3}{Q1^2}\gamma 12bar - \frac{n3}{Q1}\gamma 12bar^{1.5} + \frac{n2}{Q1}\gamma 12bar^{1.5} * \beta 22bar^{1.5} - \frac{n2}{Q1}\beta 12^{1.5} - \frac{n2\ Q3}{Q1^2}\beta 22bar^{1.5} * \gamma 12bar + \frac{n2\ Q2}{Q1^2}\gamma 12bar\ \beta 22bar\right)\Bigg) * s1;$$

$$K111 = \frac{-(\beta 12)^3}{Q2}h4 - \frac{(\gamma 12bar)^3}{Q3}h5 - \frac{1}{Q1}h6 + \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}(\gamma 12bar)^{\frac{3}{2}}s3 - \frac{n2\,Q2}{3\,\eta\,Q3^2}(\gamma 12bar)^3\,\beta 22bar\,s3 - \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\,s2 + \frac{n2\,Q2}{3\,\eta\,Q3^2}(\gamma 12bar)^3(\beta 22bar)^2 s2 + \left(\frac{n2}{3\,\eta\,Q1}(\beta 12)^{\frac{3}{2}} + \frac{n3}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{3}{2}} - \frac{n3\,Q3}{3\,\eta\,Q1^2}\gamma 12bar - \frac{n2}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{3}{2}}(\beta 22bar)^{\frac{3}{2}} - \frac{n2\,Q2}{3\,\eta\,Q1^2}\beta 22bar\,\gamma 12bar + \frac{n2\,Q3}{3\,\eta\,Q1^2}(\beta 22bar)^{\frac{3}{2}}\gamma 12bar\right)s1 - \left(\frac{n2}{3\,\eta\,Q1}(\beta 12)^{\frac{3}{2}}\gamma 12bar + \frac{n3}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{5}{2}} - \frac{n3\,Q3}{3\,\eta\,Q1^2}(\gamma 12bar)^2 - \frac{n2}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{5}{2}}(\beta 22bar)^{\frac{3}{2}} - \frac{n2\,Q2}{3\,\eta\,Q1^2}(\gamma 12bar)^2\,\beta 22bar + \frac{n2\,Q3}{3\,\eta\,Q1^2}(\beta 22bar)^{\frac{3}{2}}(\gamma 12bar)^2\right)s3 + \frac{n2}{3\,\eta}\frac{(\beta 12)^3}{Q2}s2 - \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}(\gamma 12bar)^{\frac{3}{2}}\beta 22bar\,s2 + \frac{n2}{3\,\eta\,Q3}(\beta 12)^{\frac{3}{2}}(\gamma 12bar)^{\frac{3}{2}}s3 - \frac{n2}{3\,\eta\,Q1}(\beta 12)^{\frac{3}{2}}\gamma 12bar\,s3 + \frac{n3}{3\,\eta\,Q3}(\gamma 12bar)^3 s3 - \frac{n2}{3\,\eta\,Q3}(\beta 22bar)^{\frac{3}{2}}(\gamma 12bar)^3 s3 - \frac{n3}{3\,\eta\,Q1}(\gamma 12bar)^{\frac{5}{2}}s3 + \frac{n2}{3\,\eta\,Q1}(\beta 22bar)^{\frac{3}{2}}(\gamma 12bar)^{\frac{5}{2}}s3 + \frac{n1}{3\,\eta\,Q1}s1;$$

(*K11 and K111 should be equal *)

$$T1 = c\left(\frac{4}{7}\left(alfa1 * g1\left(\beta 12^{\frac{5}{2}} - \gamma 12bar^{\frac{5}{2}} * \beta 22bar^{\frac{5}{2}}\right) + alfa2 * g2\right)\sqrt{x2}\ (u2)^{\frac{5}{2}} +$$

$$\frac{4}{7}\,\text{alfa2}*\text{g3}\sqrt{\text{x3}}\,(\text{u3})^{\frac{5}{2}}+\frac{4}{7}\,\text{alfa1}*\text{g1}\sqrt{\text{x1}}\,(\text{u1})^{\frac{5}{2}}+$$

$$\frac{5\,\text{n2}}{21\,\eta}\left(\beta 22\text{bar}^{2}\,\frac{\text{Q2}}{\text{Q3}}\left(\text{alfa1}*\text{g1}*\gamma 12\text{bar}^{\frac{5}{2}}+\text{alfa2}*\text{g3}\right)-\right.$$

$$\left.\text{alfa1}*\text{g1}*\beta 12^{\frac{5}{2}}-\text{alfa2}*\text{g3}*\beta 22\text{bar}^{\frac{5}{2}}-\text{alfa2}*\text{g2}\right)\text{s2}+$$

$$\frac{5}{21\,\eta}\left(\frac{\gamma 12\text{bar}^{2}*\text{alfa1}*\text{g1}}{\text{Q1}}\left(\text{n3}*\text{Q3}+\beta 22\text{bar}*\text{n2}*\text{Q2}-\beta 22\text{bar}^{1.5}*\text{n2}*\text{Q3}\right)-\right.$$

$$\left.\left(\text{alfa1}*\text{g1}*\gamma 12\text{bar}^{\frac{5}{2}}+\text{alfa2}*\text{g3}\right)\left(\text{n3}+\beta 22\text{bar}\,\frac{\text{n2}*\text{Q2}}{\text{Q3}}-\beta 22\text{bar}^{1.5}*\text{n2}\right)\right)\text{s3}-$$

$$\frac{5\,\text{alfa1}\,\text{g1}}{21\,\eta}\left(\text{n1}+\gamma 12\text{bar}\,\frac{\text{n3}\,\text{Q3}}{\text{Q1}}+\gamma 12\text{bar}^{1.5}\left(\beta 22\text{bar}^{1.5}*\text{n2}-\text{n3}\right)-\right.$$

$$\left.\beta 12^{1.5}*\text{n2}-\gamma 12\text{bar}*\beta 22\text{bar}^{1.5}\,\frac{\text{n2}\,\text{Q3}}{\text{Q1}}+\gamma 12\text{bar}\,\beta 22\text{bar}\,\frac{\text{n2}\,\text{Q2}}{\text{Q1}}\right)\text{s1}\Bigg);$$

$$\text{V1}=-\text{d}\left(\frac{(\text{g1}\,\text{n1}-1)}{\text{n1}}\,\text{g1}\,\text{K11}+\frac{(\text{g3}\,\text{n3}-1)}{\text{n3}}\,\text{g3}\,\text{K33}+\right.$$

$$\left.\frac{(\text{g2}\,\text{n2}-1)}{\text{n2}}\,\text{g2}\,\text{K22}+2\,\text{g1}\,\text{g3}\,\text{K31}+\frac{4}{3}\,\text{g2}\,\text{g1}\,\text{K21}+\frac{4}{3}\,\text{g2}\,\text{g3}\,\text{K32}\right);$$

```
(*These T1 and V1 are obtained
 by integrating manually which is being verified.*)
Energy6bare = T1 + V1 + 4 * P / 3 * R ^ 3 * x1 ^ 3 * B + n1 * g1 * 306 + n3 * g3 * 1553 + n2 * g2 * 1553;
Energy = Energy6bare;
Volumeenergy = 4 * P / 3 * R ^ 3 * x1 ^ 3 * B;
Energywithoutmass = T1 + V1 + 4 * P / 3 * R ^ 3 * x1 ^ 3 * B;
Print["R=  ", R];
Print["normx2 (computationally)=  ", normmx2];
Print["normx2 (by hand)=  ", normx2];
Print["normx3 (computationally)=  ", normmx3];
Print["normx3 (by hand)=  ", normx3];
Print["normx1 (computationally)=  ", normmx1];
Print["normx1 (by hand)=  ", normx1];
Print["0 to x2 (h4 computationally)=  ", h4];
Print["0 to x2 (h4 by hand)=  ", AB];
```

```
 Print["x2 to x3 (h5 computationally)=  ", h5];
 Print["x2 to x3 (h5 by hand)=  ", CD];
 Print["x2 to x3 (h5 by handdddd)=  ", CCD];
 Print["x3 to x1 (h6 computationally)=  ", h6];
 Print["x3 to x1 (h6 by hand)=  ", EF];
 Print["K22  (computationally) = ", K222];
 Print["K22  (by hand) = ", K22];
 Print["K32  (computationally) = ", K332];
 Print["K32  (by hand) = ", K32];
 Print["K21  (computationally) = ", K221];
 Print["K21  (by hand) =", K21];
 Print["K31  (computationally) = ", K331];
 Print["K31  (by hand) = ", K31];
 Print["K33  (computationally) = ", K333];
 Print["K33  (by hand) = ", K33];
 Print["K11  (computationally) = ", K111];
 Print["K11  (by hand) = ", K11];
 Print["x2= ", x2];
 Print["x3= ", x3];
 Print["x1=  ", x1];
  Print["T by hand=", T1];
 Print["V by hand=", V1];
 Print["m2=", m2];
 Print["Energy without mass = ", Energywithoutmass];
 Print["Energy=  ", Energy];
 If[n == 1, If[nloop == 1, nloop = nloop + 1; SN = 1. * 10^(-1);
   Energy11 = Energy, If[nloop == 2, nloop = nloop + 1;
    H = 10^9 + 1;
    m2 = m2B;
    If[Energy - Energy11 < 0, SN = 1., SN = -1.]]]];
If[n < 3, m2 = m2 - SN * 10^(-n); Energy = F; H = F];
If[n == 3, Energy6bare = HX;
 Print["x1=", x1];
 Print["x3=", x3];
 Print["x2=", x2];
 Print["Kinetic Energy=", T1];
 Print["Potential Energy=", V1];
 Print["Volume Energy=", Volumeenergy];
]]];

(*Code for Plotting function *)
Plot1 = Plot[Evaluate[((y2[x] /.solution2)/ x)^(3 / 2)],
```

```
  {x, .00001, x2}, (*%PlotRange→{{0,0.65},{0,79.5}},*)
  (*Solution 1 = f3(x3), solution3 = f3(x3), solution3 = f1(x1) *)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17]
    , StyleForm["(f2(x)/x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot2 = Plot[Evaluate[((β22bar * y2[x] /.solution2) / x)^(3 / 2)], {x, .00001, x2},
  (*%PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All, AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f2 bar(x)/x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot3 = Plot[Evaluate[((y3[x] /.solution3) / x)^(3 / 2)], {x, x2, x3},
  (*PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f2 bar(x)/x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot4 = Plot[Evaluate[((β12 * y2[x] /.solution2) / x)^(3 / 2)], {x, .00001, x2},
  (*%PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All, AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f1(x)/x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot5 = Plot[Evaluate[((γ12bar * y3[x] /. solution3) / x)^(3 / 2)],
  {x, x2, x3}, (*PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f1(x)/x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Plot6 = Plot[Evaluate[((y1[x] /. solution1) / x)^(3 / 2)], {x, x3, x1},
  (*PlotRange→{{0,.65},{0,79.5}},*)
  PlotRange → All,
  AxesLabel → {StyleForm["x", Italic, FontSize → 17],
    StyleForm["(f1(x)/x)^(3/2)", Italic, FontSize → 17]}, PlotStyle -> {Black}]
Show[{Plot2, Plot3}]
Show[{Plot4, Plot5, Plot6}]
```

FindMinimum: The line search decreased the step size to within the tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

FindMinimum: The line search decreased the step size to within the tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

FindMinimum: The line search decreased the step size to within the tolerance specified by AccuracyGoal and PrecisionGoal but was unable to find a sufficient decrease in the function. You may need more than MachinePrecision digits of working precision to meet these tolerances.

General: Further output of FindMinimum::lstol will be suppressed during this calculation.

```
R=  0.0207081833
```

```
normx2 (computationally)=  -0.669061099
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.640966664
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.39784424
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  7.01700442
0 to x2 (h4 by hand)=  7.01700419
x2 to x3 (h5 computationally)=  1.18170937
x2 to x3 (h5 by hand)=  1.18171048
x2 to x3 (h5 by handdddd)=  1.18170988
x3 to x1 (h6 computationally)=  1.83215591
x3 to x1 (h6 by hand)=  1.83215575
K22  (computationally) = 0.615608328
K22  (by hand) = 0.615608215
K32  (computationally) = 0.589901505
K32  (by hand) = 0.589901415
K21  (computationally) = 0.380273556
K21  (by hand) =0.380273543
K31  (computationally) = 0.377452908
K31  (by hand) = 0.377453182
K33  (computationally) = 0.569156584
K33  (by hand) = 0.569157655
K11  (computationally) = 0.328480224
K11  (by hand) = 0.32847997
x2= 0.218006159
x3= 0.236065954
x1=  0.399428363
T by hand=371.09338
V by hand=-106.899506
m2=22
Energy without mass = 581.940096
Energy=  12653.9401
R=  0.0207081833
normx2 (computationally)=  -0.669061099
normx2 (by hand)=  -0.669061195
```

```
normx3 (computationally)=  -0.640966685
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.397844231
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  7.02026646
0 to x2 (h4 by hand)=  7.02026623
x2 to x3 (h5 computationally)=  1.18226989
x2 to x3 (h5 by hand)=  1.18227124
x2 to x3 (h5 by handdddd)=  1.18227041
x3 to x1 (h6 computationally)=  1.83305027
x3 to x1 (h6 by hand)=  1.83304997
K22  (computationally) = 0.615750497
K22  (by hand) = 0.615750382
K32  (computationally) = 0.590037974
K32  (by hand) = 0.590037882
K21  (computationally) = 0.380363498
K21  (by hand) =0.380363486
K31  (computationally) = 0.377542309
K31  (by hand) = 0.377542643
K33  (computationally) = 0.569288792
K33  (by hand) = 0.569290101
K11  (computationally) = 0.328558346
K11  (by hand) = 0.328557845
x2= 0.217955478
x3= 0.236010857
x1=  0.399331701
T by hand=371.271433
V by hand=-106.92472
m2=22.01
Energy without mass = 581.862306
Energy=  12653.8623
```

FindMinimum: Working precision MachinePrecision is insufficient to achieve the requested accuracy or precision.

FindMinimum: Working precision MachinePrecision is insufficient to achieve the requested accuracy or precision.

```
R=  0.0207081833
normx2 (computationally)=  -0.66906113
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.640966663
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.397844238
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  7.04962592
0 to x2 (h4 by hand)=  7.04962588
x2 to x3 (h5 computationally)=  1.1873105
x2 to x3 (h5 by hand)=  1.18731133
x2 to x3 (h5 by handdddd)=  1.18731102
x3 to x1 (h6 computationally)=  1.84110228
x3 to x1 (h6 by hand)=  1.84110211
K22  (computationally) = 0.61702876
K22  (by hand) = 0.617028741
K32  (computationally) = 0.591264597
K32  (by hand) = 0.591264582
K21  (computationally) = 0.381172305
K21  (by hand) =0.381172303
K31  (computationally) = 0.378345264
K31  (by hand) = 0.378345472
K33  (computationally) = 0.570472993
K33  (by hand) = 0.570473824
K11  (computationally) = 0.329264318
K11  (by hand) = 0.329264027
x2= 0.217500928
x3= 0.235516643
x1=  0.398464941
T by hand=372.874308
V by hand=-107.15188
m2=22.1
Energy without mass = 581.174979
Energy=  12653.175
```

FindMinimum: Working precision MachinePrecision is insufficient to achieve the requested accuracy or precision.

General: Further output of FindMinimum::fmdig will be suppressed during this calculation.

```
R=  0.0207081833
normx2 (computationally)=  -0.669061086
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.640966663
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.397844237
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  7.08224827
0 to x2 (h4 by hand)=  7.08224796
x2 to x3 (h5 computationally)=  1.19291331
x2 to x3 (h5 by hand)=  1.19291455
x2 to x3 (h5 by handdddd)=  1.19291383
x3 to x1 (h6 computationally)=  1.85004999
x3 to x1 (h6 by hand)=  1.8500498
K22  (computationally) = 0.618445546
K22  (by hand) = 0.618445392
K32  (computationally) = 0.592624356
K32  (by hand) = 0.592624233
K21  (computationally) = 0.382068987
K21  (by hand) =0.38206897
K31  (computationally) = 0.379235835
K31  (by hand) = 0.379236138
K33  (computationally) = 0.571787324
K33  (by hand) = 0.571788498
K11  (computationally) = 0.330046681
K11  (by hand) = 0.330046392
x2= 0.216999162
x3= 0.234971127
x1=  0.397508339
T by hand=374.655457
V by hand=-107.403719
m2=22.2
Energy without mass = 580.437801
Energy=  12652.4378
R=  0.0207081833
```

normx2 (computationally)= −0.669061086

normx2 (by hand)= −0.669061195

normx3 (computationally)= −0.640966664

normx3 (by hand)= −0.640966692

normx1 (computationally)= −0.397844233

normx1 (by hand)= −0.39784424

0 to x2 (h4 computationally)= 7.11487267

0 to x2 (h4 by hand)= 7.11487235

x2 to x3 (h5 computationally)= 1.19851596

x2 to x3 (h5 by hand)= 1.19851723

x2 to x3 (h5 by handdddd)= 1.19851649

x3 to x1 (h6 computationally)= 1.85899898

x3 to x1 (h6 by hand)= 1.85899876

K22 (computationally) = 0.619859395

K22 (by hand) = 0.61985924

K32 (computationally) = 0.593981202

K32 (by hand) = 0.593981078

K21 (computationally) = 0.382963654

K21 (by hand) =0.382963637

K31 (computationally) = 0.380124256

K31 (by hand) = 0.380124565

K33 (computationally) = 0.573098204

K33 (by hand) = 0.5730994

K11 (computationally) = 0.330827177

K11 (by hand) = 0.330826825

x2= 0.216500832

x3= 0.234429349

x1= 0.396558471

T by hand=376.436794

V by hand=−107.654956

m2=22.3

Energy without mass = 579.728134

Energy= 12 651.7281

R= 0.0207081833

normx2 (computationally)= −0.669061144

normx2 (by hand)= −0.669061195

```
normx3 (computationally)=  -0.640966672
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.397844236
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  7.14749924
0 to x2 (h4 by hand)=  7.14749929
x2 to x3 (h5 computationally)=  1.20411833
x2 to x3 (h5 by hand)=  1.20411913
x2 to x3 (h5 by handdddd)=  1.20411886
x3 to x1 (h6 computationally)=  1.86794935
x3 to x1 (h6 by hand)=  1.86794912
K22  (computationally) = 0.621270418
K22  (by hand) = 0.621270442
K32  (computationally) = 0.595335212
K32  (by hand) = 0.595335232
K21  (computationally) = 0.383856337
K21  (by hand) =0.38385634
K31  (computationally) = 0.381010525
K31  (by hand) = 0.381010731
K33  (computationally) = 0.574405565
K33  (by hand) = 0.574406403
K11  (computationally) = 0.331605964
K11  (by hand) = 0.331605571
x2= 0.216005903
x3= 0.233891266
x1=  0.395615263
T by hand=378.218331
V by hand=-107.90562
m2=22.4
Energy without mass = 579.045538
Energy=  12651.0455
R=  0.0207081833
normx2 (computationally)=  -0.669061077
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.640966669
normx3 (by hand)=  -0.640966692
```

```
normx1 (computationally)=  -0.39784424
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  7.18012611
0 to x2 (h4 by hand)=  7.18012574
x2 to x3 (h5 computationally)=  1.20972296
x2 to x3 (h5 by hand)=  1.20972438
x2 to x3 (h5 by handdddd)=  1.20972349
x3 to x1 (h6 computationally)=  1.87690108
x3 to x1 (h6 by hand)=  1.87690089
K22  (computationally) = 0.622677557
K22  (by hand) = 0.622677373
K32  (computationally) = 0.596685762
K32  (by hand) = 0.596685615
K21  (computationally) = 0.384746981
K21  (by hand) =0.384746961
K31  (computationally) = 0.381895176
K31  (by hand) = 0.381895523
K33  (computationally) = 0.575711323
K33  (by hand) = 0.575712664
K11  (computationally) = 0.332383212
K11  (by hand) = 0.332382924
x2= 0.215514315
x3= 0.233356853
x1=  0.394678646
T by hand=380.000105
V by hand=-108.155801
m2=22.5
Energy without mass = 578.389548
Energy=  12650.3895
R=  0.0207081833
normx2 (computationally)=  -0.669061123
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.640966678
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.397844231
normx1 (by hand)=  -0.39784424
```

```
0 to x2 (h4 computationally)=  7.21275571
0 to x2 (h4 by hand)=  7.21275563
x2 to x3 (h5 computationally)=  1.21532651
x2 to x3 (h5 by hand)=  1.2153276
x2 to x3 (h5 by handdddd)=  1.21532705
x3 to x1 (h6 computationally)=  1.8858539
x3 to x1 (h6 by hand)=  1.88585361
K22  (computationally) = 0.624082238
K22  (by hand) = 0.624082198
K32  (computationally) = 0.598033714
K32  (by hand) = 0.598033682
K21  (computationally) = 0.385635656
K21  (by hand) =0.385635652
K31  (computationally) = 0.382777511
K31  (by hand) = 0.382777784
K33  (computationally) = 0.577013029
K33  (by hand) = 0.577014116
K11  (computationally) = 0.333158241
K11  (by hand) = 0.333157741
x2= 0.215026057
x3= 0.232826046
x1=  0.393748518
T by hand=381.782014
V by hand=-108.405301
m2=22.6
Energy without mass = 577.759783
Energy=  12649.7598
R=  0.0207081833
normx2 (computationally)=  -0.669061132
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.640966674
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.39784423
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  7.24538648
0 to x2 (h4 by hand)=  7.24538645
```

```
x2 to x3 (h5 computationally)=  1.22093091
x2 to x3 (h5 by hand)=  1.22093187
x2 to x3 (h5 by handdddd)=  1.22093145
x3 to x1 (h6 computationally)=  1.89480819
x3 to x1 (h6 by hand)=  1.8948079
K22  (computationally) = 0.625483589
K22  (by hand) = 0.625483577
K32  (computationally) = 0.599378536
K32  (by hand) = 0.599378526
K21  (computationally) = 0.386522362
K21  (by hand) =0.386522361
K31  (computationally) = 0.383657973
K31  (by hand) = 0.383658218
K33  (computationally) = 0.578312065
K33  (by hand) = 0.578313046
K11  (computationally) = 0.333931866
K11  (by hand) = 0.333931366
x2= 0.214541076
x3= 0.232298818
x1=  0.392824833
T by hand=383.564156
V by hand=-108.65431
m2=22.7
Energy without mass = 577.155802
Energy=  12 649.1558
R=  0.0207081833
normx2 (computationally)=  -0.669061144
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.640966683
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.397844231
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  7.27801882
0 to x2 (h4 by hand)=  7.27801887
x2 to x3 (h5 computationally)=  1.22653598
x2 to x3 (h5 by hand)=  1.22653693
```

```
x2 to x3 (h5 by handdddd)=  1.22653653
x3 to x1 (h6 computationally)=  1.9037637
x3 to x1 (h6 by hand)=  1.90376339
K22  (computationally) = 0.626881875
K22  (by hand) = 0.6268819
K32  (computationally) = 0.600720427
K32  (by hand) = 0.600720447
K21  (computationally) = 0.387407108
K21  (by hand) =0.38740711
K31  (computationally) = 0.38453653
K31  (by hand) = 0.384536776
K33  (computationally) = 0.579608444
K33  (by hand) = 0.579609439
K11  (computationally) = 0.334703714
K11  (by hand) = 0.334703163
x2= 0.214059339
x3= 0.231775129
x1=  0.391907506
T by hand=385.346491
V by hand=-108.902765
m2=22.8
Energy without mass = 576.577201
Energy=  12 648.5772
R=  0.0207081833
normx2 (computationally)=  -0.669061085
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.640966662
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.39784424
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  7.31065197
0 to x2 (h4 by hand)=  7.31065164
x2 to x3 (h5 computationally)=  1.2321421
x2 to x3 (h5 by hand)=  1.2321434
x2 to x3 (h5 by handdddd)=  1.23214265
x3 to x1 (h6 computationally)=  1.91272076
```

```
x3 to x1 (h6 by hand)=  1.91272058
K22  (computationally) = 0.628276673
K22  (by hand) = 0.62827651
K32  (computationally) = 0.602059088
K32  (by hand) = 0.602058958
K21  (computationally) = 0.388289906
K21  (by hand) =0.388289888
K31  (computationally) = 0.385413287
K31  (by hand) = 0.385413605
K33  (computationally) = 0.58090228
K33  (by hand) = 0.580903513
K11  (computationally) = 0.335474356
K11  (by hand) = 0.335474089
x2= 0.213580803
x3= 0.231254939
x1=  0.390996479
T by hand=387.129073
V by hand=-109.150769
m2=22.9
Energy without mass = 576.023576
Energy=  12 648.0236
R=  0.0207081833
normx2 (computationally)=  -0.669061114
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.640966678
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.39784423
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  7.34328755
0 to x2 (h4 by hand)=  7.3432874
x2 to x3 (h5 computationally)=  1.23774798
x2 to x3 (h5 by hand)=  1.23774918
x2 to x3 (h5 by handdddd)=  1.23774854
x3 to x1 (h6 computationally)=  1.92167865
x3 to x1 (h6 by hand)=  1.92167834
K22  (computationally) = 0.62966895
```

```
K22  (by hand) = 0.629668879
K32  (computationally) = 0.603395183
K32  (by hand) = 0.603395126
K21  (computationally) = 0.389170757
K21  (by hand) =0.389170749
K31  (computationally) = 0.386287942
K31  (by hand) = 0.386288243
K33  (computationally) = 0.58219291
K33  (by hand) = 0.582194099
K11  (computationally) = 0.33624249
K11  (by hand) = 0.336241965
x2= 0.21310545
x3= 0.230738206
x1=  0.390091644
T by hand=388.911763
V by hand=-109.398075
m2=23.
Energy without mass = 575.494556
Energy=  12 647.4946
R=  0.0207081833
normx2 (computationally)=  -0.669061141
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.640966662
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.39784423
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  7.37592464
0 to x2 (h4 by hand)=  7.37592468
x2 to x3 (h5 computationally)=  1.24335376
x2 to x3 (h5 by hand)=  1.24335452
x2 to x3 (h5 by handdddd)=  1.24335433
x3 to x1 (h6 computationally)=  1.9306381
x3 to x1 (h6 by hand)=  1.93063784
K22  (computationally) = 0.6310582
K22  (by hand) = 0.631058218
K32  (computationally) = 0.604728327
```

K32 (by hand) = 0.604728342

K21 (computationally) = 0.390049717

K21 (by hand) =0.390049719

K31 (computationally) = 0.387160589

K31 (by hand) = 0.387160784

K33 (computationally) = 0.583480147

K33 (by hand) = 0.583480937

K11 (computationally) = 0.337009432

K11 (by hand) = 0.337008977

x2= 0.212633234

x3= 0.230224887

x1= 0.389192968

T by hand=390.694686

V by hand=-109.644905

m2=23.1

Energy without mass = 574.989754

Energy= 12646.9898

R= 0.0207081833

normx2 (computationally)= -0.669061108

normx2 (by hand)= -0.669061195

normx3 (computationally)= -0.640966663

normx3 (by hand)= -0.640966692

normx1 (computationally)= -0.39784423

normx1 (by hand)= -0.39784424

0 to x2 (h4 computationally)= 7.40856258

0 to x2 (h4 by hand)= 7.4085624

x2 to x3 (h5 computationally)= 1.24896133

x2 to x3 (h5 by hand)= 1.24896244

x2 to x3 (h5 by handdddd)= 1.2489619

x3 to x1 (h6 computationally)= 1.9395987

x3 to x1 (h6 by hand)= 1.93959844

K22 (computationally) = 0.632444066

K22 (by hand) = 0.632443974

K32 (computationally) = 0.606058392

K32 (by hand) = 0.606058318

K21 (computationally) = 0.390926736

```
K21  (by hand) =0.390926727
K31  (computationally) = 0.388031611
K31  (by hand) = 0.388031887
K33  (computationally) = 0.584765683
K33  (by hand) = 0.584766767
K11  (computationally) = 0.337774626
K11  (by hand) = 0.337774182
x2= 0.212164115
x3= 0.229714962
x1=  0.388300369
T by hand=392.477807
V by hand=-109.891224
m2=23.2
Energy without mass = 574.50877
Energy=  12646.5088
R=  0.0207081833
normx2 (computationally)=  -0.669061142
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.640966675
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.39784423
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  7.44120275
0 to x2 (h4 by hand)=  7.44120279
x2 to x3 (h5 computationally)=  1.25456846
x2 to x3 (h5 by hand)=  1.25456937
x2 to x3 (h5 by handdddd)=  1.25456903
x3 to x1 (h6 computationally)=  1.9485605
x3 to x1 (h6 by hand)=  1.94856019
K22  (computationally) = 0.633827366
K22  (by hand) = 0.633827386
K32  (computationally) = 0.60738585
K32  (by hand) = 0.607385866
K21  (computationally) = 0.391801883
K21  (by hand) =0.391801885
K31  (computationally) = 0.388900562
```

K31 (by hand) = 0.388900796

K33 (computationally) = 0.586047805

K33 (by hand) = 0.586048753

K11 (computationally) = 0.338538024

K11 (by hand) = 0.33853749

x2= 0.211698074

x3= 0.229208379

x1= 0.387413774

T by hand=394.261097

V by hand=−110.136965

m2=23.3

Energy without mass = 574.051268

Energy= 12 646.0513

R= 0.0207081833

normx2 (computationally)= −0.669061076

normx2 (by hand)= −0.669061195

normx3 (computationally)= −0.640966674

normx3 (by hand)= −0.640966692

normx1 (computationally)= −0.39784423

normx1 (by hand)= −0.39784424

0 to x2 (h4 computationally)= 7.47384332

0 to x2 (h4 by hand)= 7.47384292

x2 to x3 (h5 computationally)= 1.26017745

x2 to x3 (h5 by hand)= 1.26017902

x2 to x3 (h5 by handdddd)= 1.26017803

x3 to x1 (h6 computationally)= 1.95752359

x3 to x1 (h6 by hand)= 1.95752329

K22 (computationally) = 0.635207079

K22 (by hand) = 0.635206882

K32 (computationally) = 0.608710081

K32 (by hand) = 0.608709924

K21 (computationally) = 0.39267511

K21 (by hand) =0.392675089

K31 (computationally) = 0.389767932

K31 (by hand) = 0.389768316

K33 (computationally) = 0.587328191

```
K33  (by hand) = 0.587329676
K11  (computationally) = 0.339299962
K11  (by hand) = 0.339299472
x2= 0.211235058
x3= 0.228705114
x1=  0.386533128
T by hand=396.044606
V by hand=-110.382242
m2=23.4
Energy without mass = 573.616857
Energy=  12 645.6169
R=  0.0207081833
normx2 (computationally)=  -0.669061092
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.64096667
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.39784423
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  7.50648625
0 to x2 (h4 by hand)=  7.50648596
x2 to x3 (h5 computationally)=  1.26578551
x2 to x3 (h5 by hand)=  1.26578688
x2 to x3 (h5 by handdddd)=  1.26578609
x3 to x1 (h6 computationally)=  1.96648793
x3 to x1 (h6 by hand)=  1.96648765
K22  (computationally) = 0.636584352
K22  (by hand) = 0.636584208
K32  (computationally) = 0.610031773
K32  (by hand) = 0.610031659
K21  (computationally) = 0.3935465
K21  (by hand) =0.393546485
K31  (computationally) = 0.390633153
K31  (by hand) = 0.39063349
K33  (computationally) = 0.588604748
K33  (by hand) = 0.58860606
K11  (computationally) = 0.340060221
```

K11 (by hand) = 0.340059746

x2= 0.210775056

x3= 0.228205113

x1= 0.385658356

T by hand=397.828284

V by hand=−110.626949

m2=23.5

Energy without mass = 573.20522

Energy= 12 645.2052

R= 0.0207081833

normx2 (computationally)= −0.669061144

normx2 (by hand)= −0.669061195

normx3 (computationally)= −0.640966674

normx3 (by hand)= −0.640966692

normx1 (computationally)= −0.397844231

normx1 (by hand)= −0.39784424

0 to x2 (h4 computationally)= 7.53913103

0 to x2 (h4 by hand)= 7.53913109

x2 to x3 (h5 computationally)= 1.2713936

x2 to x3 (h5 by hand)= 1.2713945

x2 to x3 (h5 by handdddd)= 1.27139419

x3 to x1 (h6 computationally)= 1.97545345

x3 to x1 (h6 by hand)= 1.97545315

K22 (computationally) = 0.637958915

K22 (by hand) = 0.637958942

K32 (computationally) = 0.611350794

K32 (by hand) = 0.611350815

K21 (computationally) = 0.394416041

K21 (by hand) =0.394416044

K31 (computationally) = 0.391496438

K31 (by hand) = 0.391496671

K33 (computationally) = 0.589878297

K33 (by hand) = 0.58987924

K11 (computationally) = 0.340818758

K11 (by hand) = 0.340818227

x2= 0.210318029

```
x3= 0.22770835
x1=  0.384789394
T by hand=399.612136
V by hand=-110.871109
m2=23.6
Energy without mass = 572.816
Energy=  12 644.816
R=  0.0207081833
normx2 (computationally)=  -0.669061143
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.640966666
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.397844231
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  7.57177671
0 to x2 (h4 by hand)=  7.57177676
x2 to x3 (h5 computationally)=  1.27700277
x2 to x3 (h5 by hand)=  1.27700359
x2 to x3 (h5 by handdddd)=  1.27700336
x3 to x1 (h6 computationally)=  1.98442024
x3 to x1 (h6 by hand)=  1.98441997
K22  (computationally) = 0.639330245
K22  (by hand) = 0.639330269
K32  (computationally) = 0.612666815
K32  (by hand) = 0.612666835
K21  (computationally) = 0.395283718
K21  (by hand) =0.39528372
K31  (computationally) = 0.392358018
K31  (by hand) = 0.392358229
K33  (computationally) = 0.59114957
K33  (by hand) = 0.591150428
K11  (computationally) = 0.341575816
K11  (by hand) = 0.341575328
x2= 0.209863937
x3= 0.227214794
x1=  0.383926184
```

```
T by hand=401.396191
V by hand=-111.114786
m2=23.7
Energy without mass = 572.448842
Energy=  12644.4488
R=  0.0207081833
normx2 (computationally)=  -0.669061117
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.640966662
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.39784423
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  7.60442355
0 to x2 (h4 by hand)=  7.60442343
x2 to x3 (h5 computationally)=  1.28261289
x2 to x3 (h5 by hand)=  1.28261394
x2 to x3 (h5 by handdddd)=  1.28261349
x3 to x1 (h6 computationally)=  1.99338821
x3 to x1 (h6 by hand)=  1.99338795
K22  (computationally) = 0.640698522
K22  (by hand) = 0.640698459
K32  (computationally) = 0.613979973
K32  (by hand) = 0.613979923
K21  (computationally) = 0.396149545
K21  (by hand) =0.396149539
K31  (computationally) = 0.393217876
K31  (by hand) = 0.393218138
K33  (computationally) = 0.592418564
K33  (by hand) = 0.592419597
K11  (computationally) = 0.342331252
K11  (by hand) = 0.342330793
x2= 0.20941275
x3= 0.226724412
x1=  0.383068661
T by hand=403.180434
V by hand=-111.357957
```

```
m2=23.8
Energy without mass = 572.103416
Energy=  12 644.1034
R=  0.0207081833
normx2 (computationally)=  −0.6690611
normx2 (by hand)=  −0.669061195
normx3 (computationally)=  −0.640966662
normx3 (by hand)=  −0.640966692
normx1 (computationally)=  −0.39784423
normx1 (by hand)=  −0.39784424
0 to x2 (h4 computationally)=  7.63707194
0 to x2 (h4 by hand)=  7.6370717
x2 to x3 (h5 computationally)=  1.2882234
x2 to x3 (h5 by hand)=  1.28822462
x2 to x3 (h5 by handdddd)=  1.288224
x3 to x1 (h6 computationally)=  2.00235736
x3 to x1 (h6 by hand)=  2.00235709
K22  (computationally) = 0.642063984
K22  (by hand) = 0.642063864
K32  (computationally) = 0.615290412
K32  (by hand) = 0.615290317
K21  (computationally) = 0.397013551
K21  (by hand) =0.397013538
K31  (computationally) = 0.394075907
K31  (by hand) = 0.39407621
K33  (computationally) = 0.593684855
K33  (by hand) = 0.593686036
K11  (computationally) = 0.343085053
K11  (by hand) = 0.343084596
x2= 0.208964443
x3= 0.226237166
x1=  0.38221676
T by hand=404.964855
V by hand=−111.600609
m2=23.9
Energy without mass = 571.779404
```

```
Energy=  12643.7794
R=  0.0207081833
normx2 (computationally)=  -0.669061126
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.640966685
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.39784423
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  7.66972222
0 to x2 (h4 by hand)=  7.66972215
x2 to x3 (h5 computationally)=  1.2938341
x2 to x3 (h5 by hand)=  1.29383535
x2 to x3 (h5 by handdddd)=  1.29383471
x3 to x1 (h6 computationally)=  2.01132757
x3 to x1 (h6 by hand)=  2.01132721
K22  (computationally) = 0.643426857
K22  (by hand) = 0.643426824
K32  (computationally) = 0.616598303
K32  (by hand) = 0.616598276
K21  (computationally) = 0.39787575
K21  (by hand) =0.397875746
K31  (computationally) = 0.394932086
K31  (by hand) = 0.394932402
K33  (computationally) = 0.594948427
K33  (by hand) = 0.594949687
K11  (computationally) = 0.343837037
K11  (by hand) = 0.343836413
x2= 0.208518987
x3= 0.225753025
x1=  0.381370417
T by hand=406.749437
V by hand=-111.842711
m2=24.
Energy without mass = 571.47649
Energy=  12643.4765
R=  0.0207081833
```

```
normx2 (computationally)=  -0.669061143
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.640966662
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.397844229
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  7.70237383
0 to x2 (h4 by hand)=  7.70237388
x2 to x3 (h5 computationally)=  1.29944446
x2 to x3 (h5 by hand)=  1.29944526
x2 to x3 (h5 by handdddd)=  1.29944508
x3 to x1 (h6 computationally)=  2.02029922
x3 to x1 (h6 by hand)=  2.02029894
K22  (computationally) = 0.644786838
K22  (by hand) = 0.644786861
K32  (computationally) = 0.617903365
K32  (by hand) = 0.617903384
K21  (computationally) = 0.398736165
K21  (by hand) =0.398736167
K31  (computationally) = 0.395786314
K31  (by hand) = 0.395786519
K33  (computationally) = 0.596208522
K33  (by hand) = 0.596209354
K11  (computationally) = 0.344587856
K11  (by hand) = 0.344587361
x2= 0.208076348
x3= 0.225271944
x1=  0.380529586
T by hand=408.534224
V by hand=-112.084346
m2=24.1
Energy without mass = 571.194363
Energy=  12643.1944
R=  0.0207081833
normx2 (computationally)=  -0.669061077
normx2 (by hand)=  -0.669061195
```

```
normx3 (computationally)=  -0.640966685
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.39784423
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  7.73502593
0 to x2 (h4 by hand)=  7.73502553
x2 to x3 (h5 computationally)=  1.30505759
x2 to x3 (h5 by hand)=  1.30505936
x2 to x3 (h5 by handdddd)=  1.30505821
x3 to x1 (h6 computationally)=  2.02927179
x3 to x1 (h6 by hand)=  2.02927142
K22  (computationally) = 0.646143474
K22  (by hand) = 0.646143274
K32  (computationally) = 0.619205485
K32  (by hand) = 0.619205326
K21  (computationally) = 0.399594735
K21  (by hand) =0.399594713
K31  (computationally) = 0.396639246
K31  (by hand) = 0.396639679
K33  (computationally) = 0.59746803
K33  (by hand) = 0.597469714
K11  (computationally) = 0.345336793
K11  (by hand) = 0.345336197
x2= 0.207636489
x3= 0.224793922
x1=  0.379694196
T by hand=410.319191
V by hand=-112.325494
m2=24.2
Energy without mass = 570.932679
Energy=  12642.9327
R=  0.0207081833
normx2 (computationally)=  -0.669061143
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.640966685
normx3 (by hand)=  -0.640966692
```

```
normx1 (computationally)=  -0.397844229
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  7.76768091
0 to x2 (h4 by hand)=  7.76768096
x2 to x3 (h5 computationally)=  1.31066847
x2 to x3 (h5 by hand)=  1.31066955
x2 to x3 (h5 by handdddd)=  1.31066909
x3 to x1 (h6 computationally)=  2.03824565
x3 to x1 (h6 by hand)=  2.03824528
K22  (computationally) = 0.647498144
K22  (by hand) = 0.64749817
K32  (computationally) = 0.620505369
K32  (by hand) = 0.62050539
K21  (computationally) = 0.400451589
K21  (by hand) =0.400451592
K31  (computationally) = 0.39748985
K31  (by hand) = 0.397490128
K33  (computationally) = 0.598722743
K33  (by hand) = 0.598723868
K11  (computationally) = 0.346084244
K11  (by hand) = 0.346083598
x2= 0.207199402
x3= 0.224318887
x1=  0.378864189
T by hand=412.104312
V by hand=-112.56608
m2=24.3
Energy without mass = 570.691202
Energy=  12642.6912
R=  0.0207081833
normx2 (computationally)=  -0.669061143
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.640966683
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.39784423
normx1 (by hand)=  -0.39784424
```

```
0 to x2 (h4 computationally)=  7.80033654
0 to x2 (h4 by hand)=  7.8003366
x2 to x3 (h5 computationally)=  1.31628087
x2 to x3 (h5 by hand)=  1.31628194
x2 to x3 (h5 by handdddd)=  1.3162815
x3 to x1 (h6 computationally)=  2.04722068
x3 to x1 (h6 by hand)=  2.04722032
K22  (computationally) = 0.648849612
K22  (by hand) = 0.64884964
K32  (computationally) = 0.621802334
K32  (by hand) = 0.621802356
K21  (computationally) = 0.40130665
K21  (by hand) =0.401306653
K31  (computationally) = 0.398338925
K31  (by hand) = 0.398339199
K33  (computationally) = 0.599975702
K33  (by hand) = 0.599976811
K11  (computationally) = 0.346830235
K11  (by hand) = 0.346829597
x2= 0.206765039
x3= 0.223846834
x1=  0.378039516
T by hand=413.889629
V by hand=-112.806211
m2=24.4
Energy without mass = 570.469583
Energy=  12642.4696
R=  0.0207081833
normx2 (computationally)=  -0.669061074
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.640966668
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.39784423
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  7.83299277
0 to x2 (h4 by hand)=  7.83299235
```

```
x2 to x3 (h5 computationally)=  1.32189464
x2 to x3 (h5 by hand)=  1.32189626
x2 to x3 (h5 by handdddd)=  1.32189527
x3 to x1 (h6 computationally)=  2.05619695
x3 to x1 (h6 by hand)=  2.05619664
K22  (computationally) = 0.65019786
K22  (by hand) = 0.650197647
K32  (computationally) = 0.623096356
K32  (by hand) = 0.623096186
K21  (computationally) = 0.40215993
K21  (by hand) =0.402159907
K31  (computationally) = 0.399186448
K31  (by hand) = 0.399186844
K33  (computationally) = 0.601226742
K33  (by hand) = 0.60122827
K11  (computationally) = 0.347574864
K11  (by hand) = 0.347574362
x2= 0.20633337
x3= 0.22337773
x1=  0.37722012
T by hand=415.675147
V by hand=-113.045899
m2=24.5
Energy without mass = 570.267534
Energy=  12642.2675
R=  0.0207081833
normx2 (computationally)=  -0.669061124
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.640966666
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.39784423
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  7.86565174
0 to x2 (h4 by hand)=  7.86565166
x2 to x3 (h5 computationally)=  1.32750705
x2 to x3 (h5 by hand)=  1.32750813
```

```
x2 to x3 (h5 by handdddd)=  1.32750769
x3 to x1 (h6 computationally)=  2.06517428
x3 to x1 (h6 by hand)=  2.06517398
K22  (computationally) = 0.651544128
K22  (by hand) = 0.651544089
K32  (computationally) = 0.62438821
K32  (by hand) = 0.624388178
K21  (computationally) = 0.403011507
K21  (by hand) =0.403011503
K31  (computationally) = 0.400031866
K31  (by hand) = 0.400032138
K33  (computationally) = 0.602473924
K33  (by hand) = 0.602475006
K11  (computationally) = 0.348317751
K11  (by hand) = 0.348317228
x2= 0.205904388
x3= 0.22291153
x1=  0.376405934
T by hand=417.460803
V by hand=-113.285018
m2=24.6
Energy without mass = 570.084811
Energy=  12642.0848
R=  0.0207081833
normx2 (computationally)=  -0.669061143
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.640966661
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.39784423
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  7.89831172
0 to x2 (h4 by hand)=  7.89831178
x2 to x3 (h5 computationally)=  1.33312036
x2 to x3 (h5 by hand)=  1.33312118
x2 to x3 (h5 by handdddd)=  1.333121
x3 to x1 (h6 computationally)=  2.07415278
```

```
x3 to x1 (h6 by hand)=  2.0741525
K22  (computationally) = 0.652887462
K22  (by hand) = 0.652887489
K32  (computationally) = 0.625677316
K32  (by hand) = 0.625677338
K21  (computationally) = 0.403861338
K21  (by hand) =0.403861341
K31  (computationally) = 0.400875656
K31  (by hand) = 0.400875867
K33  (computationally) = 0.603718904
K33  (by hand) = 0.603719762
K11  (computationally) = 0.349059181
K11  (by hand) = 0.34905868
x2= 0.20547805
x3= 0.222448216
x1=  0.375596912
T by hand=419.246643
V by hand=-113.52367
m2=24.7
Energy without mass = 569.921101
Energy=  12641.9211
R=  0.0207081833
normx2 (computationally)=  -0.669061113
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.640966665
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.39784423
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  7.9309725
0 to x2 (h4 by hand)=  7.93097234
x2 to x3 (h5 computationally)=  1.33873512
x2 to x3 (h5 by hand)=  1.33873631
x2 to x3 (h5 by handdddd)=  1.33873576
x3 to x1 (h6 computationally)=  2.08313236
x3 to x1 (h6 by hand)=  2.08313206
K22  (computationally) = 0.654227756
```

K22 (by hand) = 0.654227677

K32 (computationally) = 0.626963639

K32 (by hand) = 0.626963576

K21 (computationally) = 0.404709417

K21 (by hand) =0.404709409

K31 (computationally) = 0.401717945

K31 (by hand) = 0.401718243

K33 (computationally) = 0.604962205

K33 (by hand) = 0.60496338

K11 (computationally) = 0.349799065

K11 (by hand) = 0.349798558

x2= 0.205054327

x3= 0.221987766

x1= 0.374792999

T by hand=421.032664

V by hand=−113.761859

m2=24.8

Energy without mass = 569.776122

Energy= 12 641.7761

R= 0.0207081833

normx2 (computationally)= −0.669061104

normx2 (by hand)= −0.669061195

normx3 (computationally)= −0.640966678

normx3 (by hand)= −0.640966692

normx1 (computationally)= −0.39784423

normx1 (by hand)= −0.39784424

0 to x2 (h4 computationally)= 7.96363488

0 to x2 (h4 by hand)= 7.96363466

x2 to x3 (h5 computationally)= 1.34435013

x2 to x3 (h5 by hand)= 1.34435159

x2 to x3 (h5 by handdddd)= 1.34435078

x3 to x1 (h6 computationally)= 2.092113

x3 to x1 (h6 by hand)= 2.09211265

K22 (computationally) = 0.65556549

K22 (by hand) = 0.655565381

K32 (computationally) = 0.628247467

```
K32  (by hand) = 0.62824738
K21  (computationally) = 0.405555786
K21  (by hand) =0.405555774
K31  (computationally) = 0.402558495
K31  (by hand) = 0.402558858
K33  (computationally) = 0.606202916
K33  (by hand) = 0.606204344
K11  (computationally) = 0.350537338
K11  (by hand) = 0.350536745
x2= 0.204633199
x3= 0.22153014
x1=  0.373994138
T by hand=422.818846
V by hand=-113.999545
m2=24.9
Energy without mass = 569.649624
Energy=  12 641.6496
R=  0.0207081833
normx2 (computationally)=  -0.669061143
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.640966671
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.39784423
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  7.99629915
0 to x2 (h4 by hand)=  7.99629921
x2 to x3 (h5 computationally)=  1.34996431
x2 to x3 (h5 by hand)=  1.34996526
x2 to x3 (h5 by handdddd)=  1.34996496
x3 to x1 (h6 computationally)=  2.10109488
x3 to x1 (h6 by hand)=  2.10109455
K22  (computationally) = 0.656900849
K22  (by hand) = 0.656900877
K32  (computationally) = 0.629528867
K32  (by hand) = 0.629528889
K21  (computationally) = 0.40640048
```

```
K21  (by hand) =0.406400483
K31  (computationally) = 0.403397091
K31  (by hand) = 0.403397337
K33  (computationally) = 0.607440057
K33  (by hand) = 0.607441052
K11  (computationally) = 0.351274261
K11  (by hand) = 0.351273684
x2= 0.204214643
x3= 0.221075301
x1=  0.373200282
T by hand=424.605196
V by hand=-114.23674
m2=25.
Energy without mass = 569.541354
Energy=  12641.5414
R=  0.0207081833
normx2 (computationally)=  -0.669061143
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.640966661
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.39784423
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  8.02896431
0 to x2 (h4 by hand)=  8.02896437
x2 to x3 (h5 computationally)=  1.35557955
x2 to x3 (h5 by hand)=  1.35558039
x2 to x3 (h5 by handdddd)=  1.35558021
x3 to x1 (h6 computationally)=  2.11007788
x3 to x1 (h6 by hand)=  2.11007759
K22  (computationally) = 0.658233282
K22  (by hand) = 0.658233309
K32  (computationally) = 0.630807551
K32  (by hand) = 0.630807572
K21  (computationally) = 0.407243464
K21  (by hand) =0.407243467
K31  (computationally) = 0.404234138
```

```
K31  (by hand) = 0.404234355
K33  (computationally) = 0.608675208
K33  (by hand) = 0.608676089
K11  (computationally) = 0.352009759
K11  (by hand) = 0.352009236
x2= 0.203798623
x3= 0.220623235
x1=  0.372411379
T by hand=426.391728
V by hand=-114.473483
m2=25.1
Energy without mass = 569.451028
Energy=  12 641.451
R=  0.0207081833
normx2 (computationally)=  -0.669061089
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.640966665
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.39784423
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  8.06163017
0 to x2 (h4 by hand)=  8.06162983
x2 to x3 (h5 computationally)=  1.36119644
x2 to x3 (h5 by hand)=  1.36119794
x2 to x3 (h5 by handdddd)=  1.36119711
x3 to x1 (h6 computationally)=  2.11906191
x3 to x1 (h6 by hand)=  2.1190616
K22  (computationally) = 0.659562703
K22  (by hand) = 0.659562537
K32  (computationally) = 0.632083499
K32  (by hand) = 0.632083366
K21  (computationally) = 0.408084734
K21  (by hand) =0.408084716
K31  (computationally) = 0.405069771
K31  (by hand) = 0.405070138
K33  (computationally) = 0.609908929
```

K33 (by hand) = 0.609910357

K11 (computationally) = 0.352743712

K11 (by hand) = 0.352743191

x2= 0.203385114

x3= 0.220173919

x1= 0.371627375

T by hand=428.178437

V by hand=−114.709777

m2=25.2

Energy without mass = 569.378385

Energy= 12 641.3784

R= 0.0207081833

normx2 (computationally)= −0.669061073

normx2 (by hand)= −0.669061195

normx3 (computationally)= −0.640966666

normx3 (by hand)= −0.640966692

normx1 (computationally)= −0.39784423

normx1 (by hand)= −0.39784424

0 to x2 (h4 computationally)= 8.09429779

0 to x2 (h4 by hand)= 8.09429734

x2 to x3 (h5 computationally)= 1.36681301

x2 to x3 (h5 by hand)= 1.3668147

x2 to x3 (h5 by handdddd)= 1.36681368

x3 to x1 (h6 computationally)= 2.12804706

x3 to x1 (h6 by hand)= 2.12804674

K22 (computationally) = 0.66088974

K22 (by hand) = 0.660889515

K32 (computationally) = 0.633357059

K32 (by hand) = 0.63335688

K21 (computationally) = 0.408924348

K21 (by hand) =0.408924324

K31 (computationally) = 0.405903586

K31 (by hand) = 0.405903998

K33 (computationally) = 0.611139611

K33 (by hand) = 0.611141201

K11 (computationally) = 0.3534762

```
K11  (by hand) = 0.353475679
x2= 0.202974099
x3= 0.219727311
x1=  0.370848221
T by hand=429.965305
V by hand=-114.94558
m2=25.3
Energy without mass = 569.3232
Energy=  12 641.3232
R=  0.0207081833
normx2 (computationally)=  -0.669061143
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.640966667
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.39784423
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  8.12696774
0 to x2 (h4 by hand)=  8.1269678
x2 to x3 (h5 computationally)=  1.37242846
x2 to x3 (h5 by hand)=  1.3724294
x2 to x3 (h5 by handdddd)=  1.37242914
x3 to x1 (h6 computationally)=  2.1370333
x3 to x1 (h6 by hand)=  2.13703297
K22  (computationally) = 0.662214732
K22  (by hand) = 0.662214761
K32  (computationally) = 0.634628443
K32  (by hand) = 0.634628467
K21  (computationally) = 0.409762336
K21  (by hand) =0.40976234
K31  (computationally) = 0.40673543
K31  (by hand) = 0.406735672
K33  (computationally) = 0.612366682
K33  (by hand) = 0.612367664
K11  (computationally) = 0.354207147
K11  (by hand) = 0.354206563
x2= 0.202565558
```

x3= 0.219283379

x1= 0.370073864

T by hand=431.752316

V by hand=−115.180862

m2=25.4

Energy without mass = 569.285242

Energy= 12 641.2852

R= 0.0207081833

normx2 (computationally)= −0.669061072

normx2 (by hand)= −0.669061195

normx3 (computationally)= −0.640966668

normx3 (by hand)= −0.640966692

normx1 (computationally)= −0.397844229

normx1 (by hand)= −0.39784424

0 to x2 (h4 computationally)= 8.15963733

0 to x2 (h4 by hand)= 8.15963687

x2 to x3 (h5 computationally)= 1.37804684

x2 to x3 (h5 by hand)= 1.37804858

x2 to x3 (h5 by handdddd)= 1.37804752

x3 to x1 (h6 computationally)= 2.14602062

x3 to x1 (h6 by hand)= 2.14602028

K22 (computationally) = 0.663536172

K22 (by hand) = 0.663535944

K32 (computationally) = 0.635896768

K32 (by hand) = 0.635896587

K21 (computationally) = 0.410598607

K21 (by hand) =0.410598582

K31 (computationally) = 0.407566155

K31 (by hand) = 0.40756658

K33 (computationally) = 0.61359328

K33 (by hand) = 0.613594924

K11 (computationally) = 0.354936773

K11 (by hand) = 0.354936217

x2= 0.202159443

x3= 0.218842123

x1= 0.369304263

```
T by hand=433.539531
V by hand=-115.415765
m2=25.5
Energy without mass = 569.264211
Energy=  12641.2642
R=  0.0207081833
normx2 (computationally)=  -0.669061143
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.640966669
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.397844229
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  8.1923099
0 to x2 (h4 by hand)=  8.19230996
x2 to x3 (h5 computationally)=  1.38366306
x2 to x3 (h5 by hand)=  1.38366403
x2 to x3 (h5 by handdddd)=  1.38366375
x3 to x1 (h6 computationally)=  2.15500903
x3 to x1 (h6 by hand)=  2.15500869
K22  (computationally) = 0.664855973
K22  (by hand) = 0.664856003
K32  (computationally) = 0.637163166
K32  (by hand) = 0.63716319
K21  (computationally) = 0.411433297
K21  (by hand) =0.411433301
K31  (computationally) = 0.408394715
K31  (by hand) = 0.408394963
K33  (computationally) = 0.614815491
K33  (by hand) = 0.6148165
K11  (computationally) = 0.355664852
K11  (by hand) = 0.355664242
x2= 0.20175576
x3= 0.218403484
x1=  0.368539361
T by hand=435.326874
V by hand=-115.650121
```

m2=25.6

Energy without mass = 569.259943

Energy= 12 641.2599

R= 0.0207081833

normx2 (computationally)= −0.669061099

normx2 (by hand)= −0.669061195

normx3 (computationally)= −0.640966668

normx3 (by hand)= −0.640966692

normx1 (computationally)= −0.39784423

normx1 (by hand)= −0.39784424

0 to x2 (h4 computationally)= 8.22498238

0 to x2 (h4 by hand)= 8.22498211

x2 to x3 (h5 computationally)= 1.38928176

x2 to x3 (h5 by hand)= 1.38928323

x2 to x3 (h5 by handdddd)= 1.38928245

x3 to x1 (h6 computationally)= 2.16399853

x3 to x1 (h6 by hand)= 2.16399819

K22 (computationally) = 0.666172419

K22 (by hand) = 0.666172286

K32 (computationally) = 0.638426631

K32 (by hand) = 0.638426525

K21 (computationally) = 0.412266303

K21 (by hand) =0.412266288

K31 (computationally) = 0.409222083

K31 (by hand) = 0.409222446

K33 (computationally) = 0.61603691

K33 (by hand) = 0.616038331

K11 (computationally) = 0.356391635

K11 (by hand) = 0.356391066

x2= 0.201354458

x3= 0.217967464

x1= 0.367779121

T by hand=437.114414

V by hand=−115.884095

m2=25.7

Energy without mass = 569.272138

```
Energy=  12 641.2721
R=  0.0207081833
normx2 (computationally)=  −0.669061143
normx2 (by hand)=  −0.669061195
normx3 (computationally)=  −0.640966669
normx3 (by hand)=  −0.640966692
normx1 (computationally)=  −0.397844229
normx1 (by hand)=  −0.39784424
0 to x2 (h4 computationally)=  8.19557714
0 to x2 (h4 by hand)=  8.1955772
x2 to x3 (h5 computationally)=  1.38422484
x2 to x3 (h5 by hand)=  1.38422581
x2 to x3 (h5 by handdddd)=  1.38422553
x3 to x1 (h6 computationally)=  2.15590793
x3 to x1 (h6 by hand)=  2.15590759
K22  (computationally) = 0.664987761
K22  (by hand) = 0.66498779
K32  (computationally) = 0.63728964
K32  (by hand) = 0.637289663
K21  (computationally) = 0.411516673
K21  (by hand) =0.411516676
K31  (computationally) = 0.408477509
K31  (by hand) = 0.408477759
K33  (computationally) = 0.614937683
K33  (by hand) = 0.614938696
K11  (computationally) = 0.355737588
K11  (by hand) = 0.355736978
x2= 0.201715523
x3= 0.218359764
x1=  0.368463128
T by hand=435.505619
V by hand=−115.673536
m2=25.61
Energy without mass = 569.260425
Energy=  12 641.2604
R=  0.0207081833
```

```
normx2 (computationally)=  -0.669061136
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.640966669
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.397844229
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  8.19884432
0 to x2 (h4 by hand)=  8.19884433
x2 to x3 (h5 computationally)=  1.38478673
x2 to x3 (h5 by hand)=  1.38478777
x2 to x3 (h5 by handdddd)=  1.38478742
x3 to x1 (h6 computationally)=  2.15680685
x3 to x1 (h6 by hand)=  2.1568065
K22  (computationally) = 0.665119483
K22  (by hand) = 0.665119488
K32  (computationally) = 0.637416065
K32  (by hand) = 0.63741607
K21  (computationally) = 0.41160003
K21  (by hand) =0.411600031
K31  (computationally) = 0.40856031
K31  (by hand) = 0.408560576
K33  (computationally) = 0.615059935
K33  (by hand) = 0.615061006
K11  (computationally) = 0.35581032
K11  (by hand) = 0.355809717
x2= 0.20167531
x3= 0.218316072
x1=  0.368386942
T by hand=435.684368
V by hand=-115.696951
m2=25.62
Energy without mass = 569.26107
Energy=  12641.2611
R=  0.0207081833
normx2 (computationally)=  -0.669061143
normx2 (by hand)=  -0.669061195
```

```
normx3 (computationally)=  -0.640966669
normx3 (by hand)=  -0.640966692
normx1 (computationally)=  -0.397844229
normx1 (by hand)=  -0.39784424
0 to x2 (h4 computationally)=  8.19590387
0 to x2 (h4 by hand)=  8.19590393
x2 to x3 (h5 computationally)=  1.38428101
x2 to x3 (h5 by hand)=  1.38428198
x2 to x3 (h5 by handdddd)=  1.3842817
x3 to x1 (h6 computationally)=  2.15599782
x3 to x1 (h6 by hand)=  2.15599748
K22  (computationally) = 0.665000942
K22  (by hand) = 0.665000971
K32  (computationally) = 0.637302288
K32  (by hand) = 0.637302312
K21  (computationally) = 0.41152501
K21  (by hand) =0.411525013
K31  (computationally) = 0.408485786
K31  (by hand) = 0.408486035
K33  (computationally) = 0.614949895
K33  (by hand) = 0.614950905
K11  (computationally) = 0.35574486
K11  (by hand) = 0.35574425
x2= 0.201711501
x3= 0.218355394
x1=  0.368455508
T by hand=435.523494
V by hand=-115.675877
m2=25.611
Energy without mass = 569.260482
Energy=  12641.2605
R=  0.0207081833
normx2 (computationally)=  -0.669061143
normx2 (by hand)=  -0.669061195
normx3 (computationally)=  -0.640966669
normx3 (by hand)=  -0.640966692
```

normx1 (computationally)= −0.397844229

normx1 (by hand)= −0.39784424

0 to x2 (h4 computationally)= 8.19623059

0 to x2 (h4 by hand)= 8.19623065

x2 to x3 (h5 computationally)= 1.38433719

x2 to x3 (h5 by hand)= 1.38433816

x2 to x3 (h5 by handdddd)= 1.38433788

x3 to x1 (h6 computationally)= 2.15608771

x3 to x1 (h6 by hand)= 2.15608737

K22 (computationally) = 0.665014117

K22 (by hand) = 0.665014145

K32 (computationally) = 0.637314932

K32 (by hand) = 0.637314955

K21 (computationally) = 0.411533346

K21 (by hand) =0.411533349

K31 (computationally) = 0.408494065

K31 (by hand) = 0.408494315

K33 (computationally) = 0.614962117

K33 (by hand) = 0.61496313

K11 (computationally) = 0.355752133

K11 (by hand) = 0.355751524

x2= 0.201707479

x3= 0.218351024

x1= 0.368447887

T by hand=435.541368

V by hand=−115.678219

m2=25.612

Energy without mass = 569.260541

Energy= 12 641.2605

x1=0.368447887

x3=0.218351024

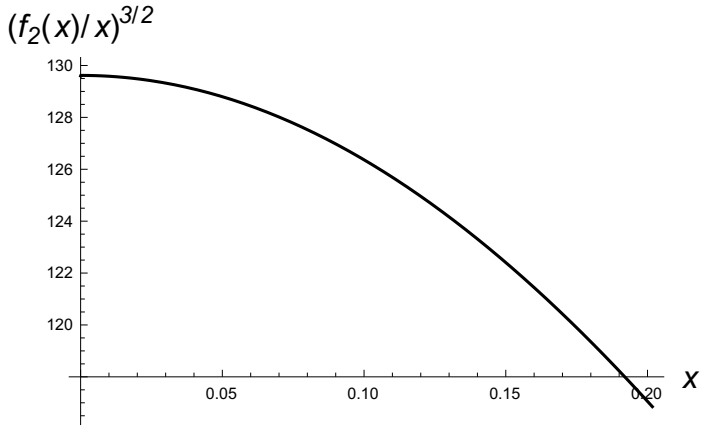
x2=0.201707479

Kinetic Energy=435.541368

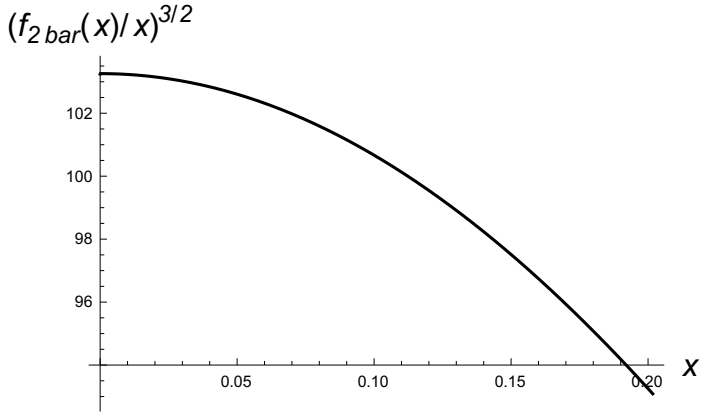
Potential Energy=−115.678219

Volume Energy=249.397392

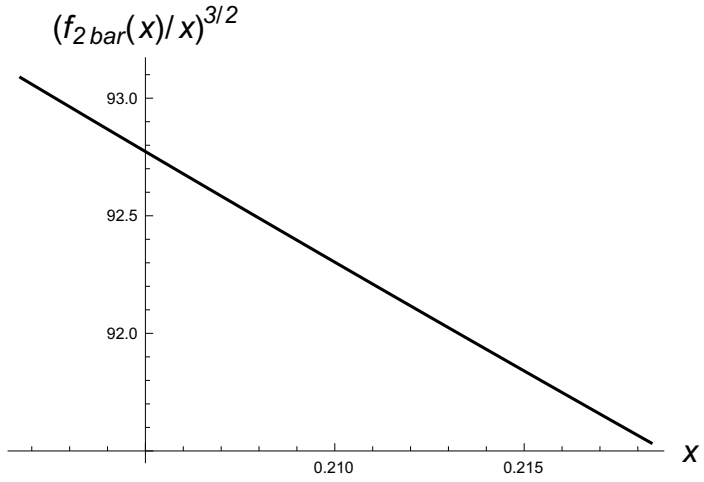
Out[ ]=


$(f_2(x)/x)^{3/2}$
130
128
126
124
122
120
0.05
0.10
0.15
0.20
x

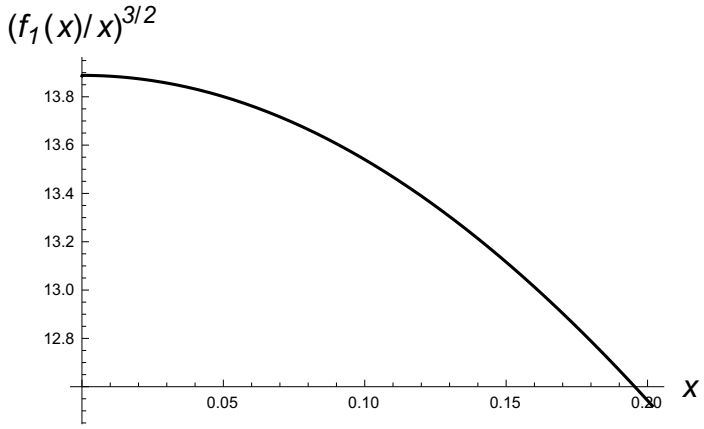
Out[ ]=


$(f_{2\,bar}(x)/x)^{3/2}$
102
100
98
96
0.05
0.10
0.15
0.20
x

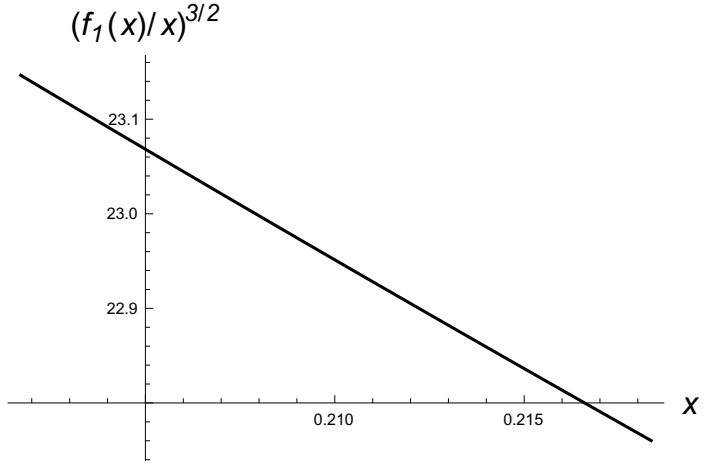
Out[ ]=


$(f_{2\,bar}(x)/x)^{3/2}$
93.0
92.5
92.0
0.210
0.215
x

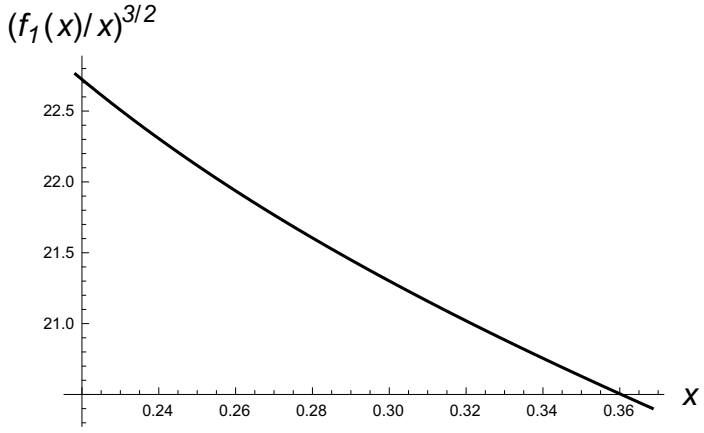
Out[ ]=


$(f_1(x)/x)^{3/2}$
13.8
13.6
13.4
13.2
13.0
12.8
0.05
0.10
0.15
0.20
x

Out[ ]=

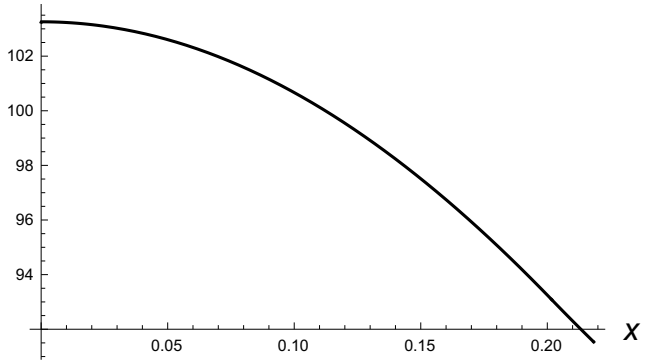

$(f_1(x)/x)^{3/2}$
23.1
23.0
22.9
0.210
0.215
x

Out[ ]=


$(f_1(x)/x)^{3/2}$
22.5
22.0
21.5
21.0
0.24
0.26
0.28
0.30
0.32
0.34
0.36
x

$(f_{2\,bar}(x)/x)^{3/2}$

Out[ ]=



$(f_1(x)/x)^{3/2}$

Out[ ]=

